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nephrons which gradually decreases the function of kidneys.

Females are prone to recurring urinary tract infections as they have shorter urethras. With age prostate in males may enlarge which forces urethra to tighten restricting a normal urinary flow.

Uremia - Uremia is characterized by increase in urea and other non-protein nitrogenous substances like uric acid and creatinine in blood. Normal urea level in human blood is about 17-30mg/100mL of blood. The urea concentration rises as 10 times of normal levels during chronic renal failure.

Renal calculi- Kidney stone or calculi, also called renal stone or nephrolithiasis, is the formation of hard stone like masses in the renal tubules of renal pelvis. It is mainly due to the accumulation of soluble crystals of salts of sodium oxalates and certain phosphates. This result in severe pain called "renal colic pain" and can cause scars in the kidneys. Renal stones can be removed by techniques like pyleothotomy or lithotripsy.

Glomerulonephritis- It is also called Bright's disease and is characterized by inflammation of the glomeruli of both kidneys and is usually due to post-streptococcal infection that occurs in children. Symptoms are haematuria, proteinuria, salt and water retention, oligouria, hypertension and pulmonary oedema.

8.8 Haemodialysis

Malfunctiong of the kidneys can lead to accumalation of urea and other toxic substances, leading to kidney failure. In such patients toxic urea can be removed from the blood by a process called haemodialysis. A dialyzing machine or an artificial kidney is connected to the patient's body. A dialyzing machine consists of a long cellulose tube surrounded by the dialysing fluid in a water bath. The patient's blood is drawn from a conveinent artery and pumped into the dialysing unit after adding an anticoagulant like heparin. The tiny pores in the dialysis tube allows small molecules such as glucose, salts and

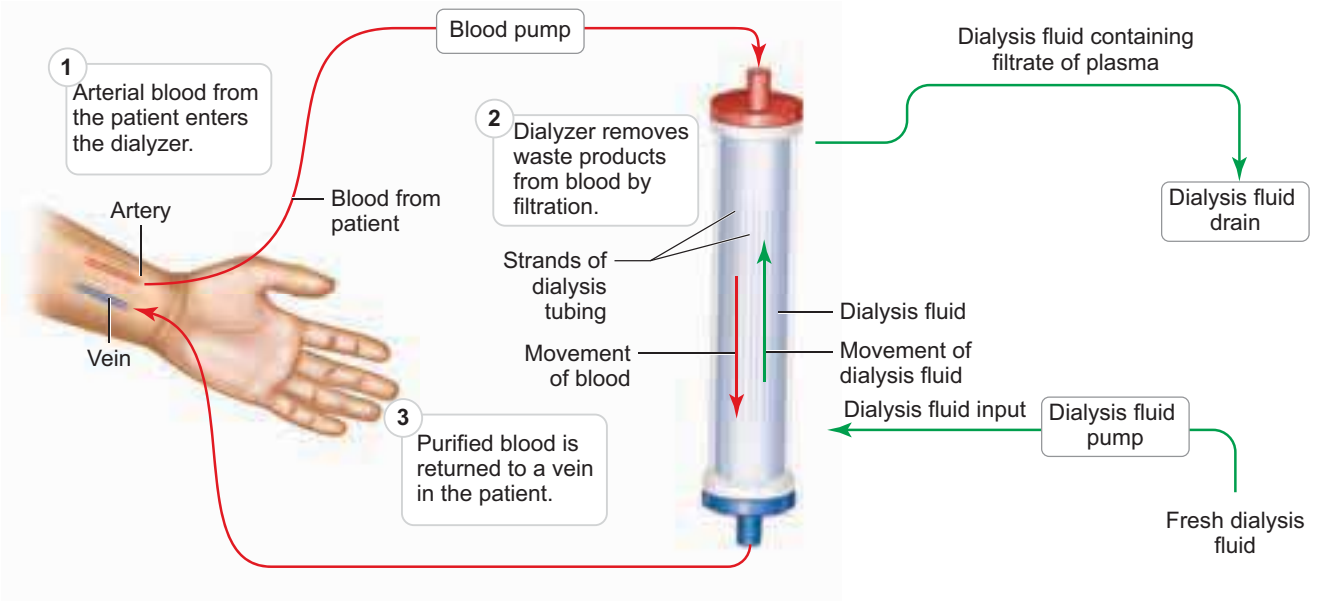

Figure 8.12 Simplified diagram of hemodialysis